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Experiment 1

Aim:
Implement Linear and Logistic Regression on real-world datasets

Theory:

Linear Regression and Logistic Regression are basic supervised machine learning techniques used to analyze relationships in data and make predictions. They are widely applied to real-world datasets in fields like business, healthcare, finance, and education.

Linear Regression

Linear Regression is used when the output to be predicted is a continuous value such as price, marks, temperature, or sales. It studies how one or more input variables influence an output variable. By learning patterns from historical data, the model predicts future values.

When implemented on real-world datasets, the algorithm first learns from known data and then predicts unknown values. For example, it can estimate house prices based on size and location or predict company sales based on advertising spend. The main goal is to understand the relationship between variables and make accurate numerical predictions.

Logistic Regression

Logistic Regression is used for classification tasks where the output belongs to categories, such as yes/no, true/false, or spam/not spam. Instead of predicting a number, it predicts which class a data point belongs to.

When applied to real-world datasets, it analyzes patterns in the data and assigns each input to a category. For instance, it can predict whether a student will pass or fail, whether a customer will buy a product, or whether an email is spam. It helps in decision-making by estimating the likelihood of an event occurring.

Implementation:

Read Data

```
[7]
0s
df = pd.read_csv('train.csv')
```

Clean Data

```
important = df[['Survived','Pclass','Age','SibSp','Parch','Fare']]
print(important.head())
```

```
   Survived  Pclass   Age  SibSp  Parch     Fare
0         0       3  22.0      1      0   7.2500
1         1       1  38.0      1      0  71.2833
2         1       3  26.0      0      0   7.9250
3         1       1  35.0      1      0  53.1000
4         0       3  35.0      0      0   8.0500
```

```
print("Missing values:")
print(important.isnull().sum())
```

```
Missing values:
Survived      0
Pclass        0
Age         177
SibSp         0
Parch         0
Fare          0
dtype: int64
```

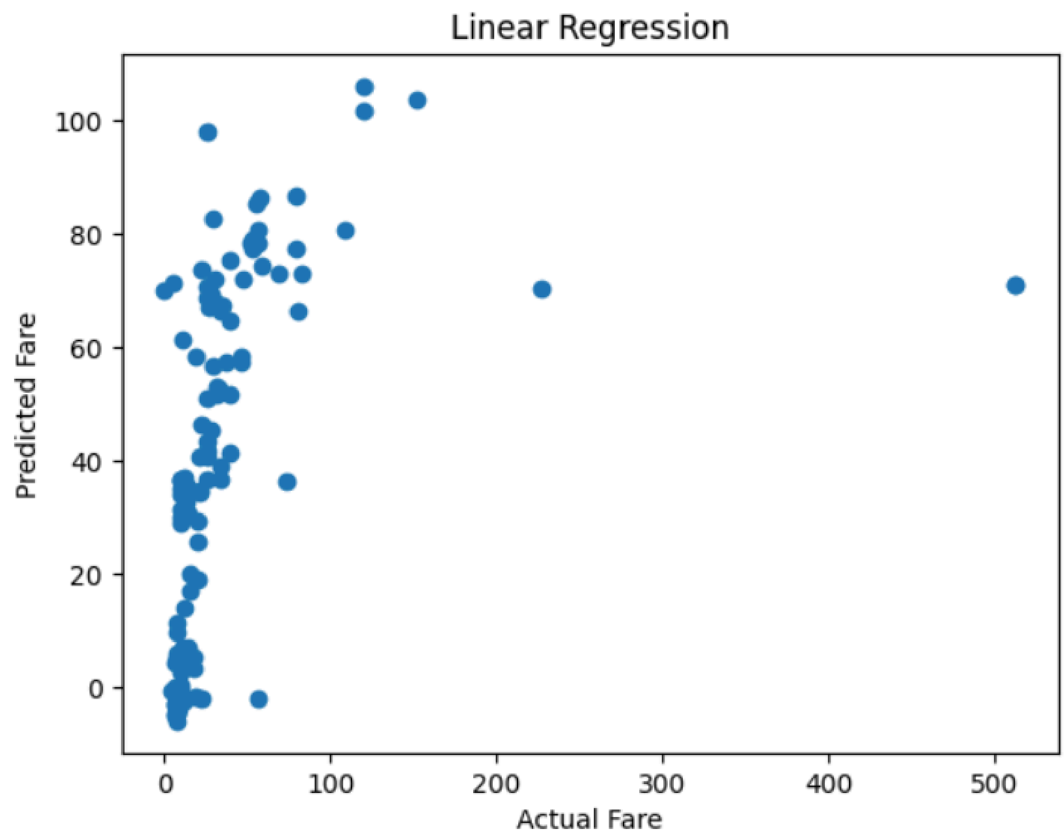
Linear regression

```
X_lin = df[['Pclass','Age','SibSp','Parch']]
y_lin = df['Fare']

X_train_lin, X_test_lin, y_train_lin, y_test_lin = train_test_split(
    X_lin, y_lin, test_size=0.2, random_state=42
)
```

```
lin = LinearRegression()
lin.fit(X_train_lin, y_train_lin)
pred_lin = lin.predict(X_test_lin)

print("LINEAR REGRESSION")
print("MSE:", mean_squared_error(y_test_lin, pred_lin))
print("R2:", r2_score(y_test_lin, pred_lin))
```

```
LINEAR REGRESSION
MSE: 3424.5685181617578
R2: 0.177436773223145
```

...

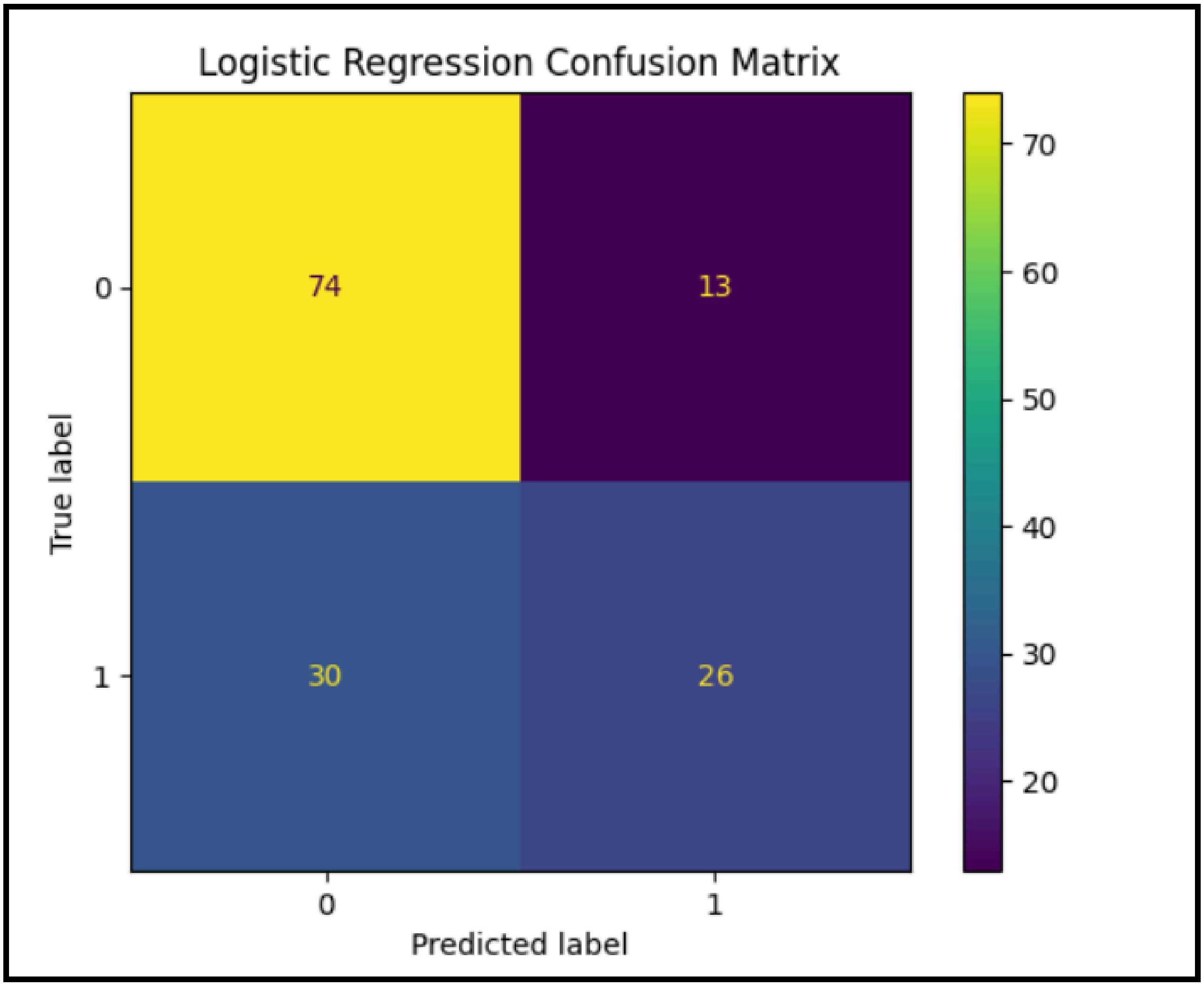


Logistic regression

```
X_log = df[['Pclass','Age','SibSp','Parch','Fare']]
y_log = df['Survived']

X_train_log, X_test_log, y_train_log, y_test_log = train_test_split(
    X_log, y_log, test_size=0.2, random_state=42
)
```

```
log = LogisticRegression(max_iter=5000)
log.fit(X_train_log, y_train_log)
pred_log = log.predict(X_test_log)

print("LOGISTIC REGRESSION")
print("Accuracy:", accuracy_score(y_test_log, pred_log))
```

```
LOGISTIC REGRESSION
Accuracy: 0.6993006993006993
```



Conclusion:

Linear regression and logistic regression were implemented on the Titanic dataset to predict fare and survival. This experiment shows that linear regression is used for numerical prediction, while logistic regression is used for classification tasks.